Name _____ _____ Baggett/Fall2026/207/Quiz6

first name last name

No Calculators.

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 1 for the reason.**

1. (**2** pts) In a related rates problem you differentiate the relating equation with respect to time *before* substituting the given numerical values.

True **False**

Why? **Substituting first freezes the variables, and a constant has derivative zero – the relationship you needed is destroyed.**

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2. (**2** pts) $\dfrac{d}{dx}(\ln 10) = \dfrac{1}{10}$

True **False**

Why? $\ln 10$ **is a constant, so the derivative is** 0**. The rule** $\frac{d}{dx}\ln x = \frac{1}{x}$ **needs a variable inside.**

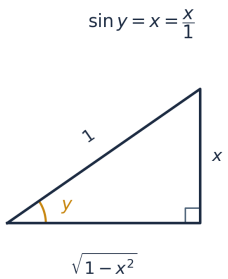
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3. (**6** pts) **Derive** $\dfrac{d}{dx}[\arcsin x]$ using implicit differentiation. **Show every step.**

You may use the derivative rules for sin and cos, but **not** any derivative rule for an inverse trig function. Quoting the formula earns nothing.

$$\frac{d}{dx}[\arcsin x] = \frac{1}{\sqrt{1-x^2}}$$

$y = \arcsin x$	name it
$\sin y = x$	**rewrite** — no inverse left in it
$\cos y \dfrac{dy}{dx} = 1$	**differentiate** both sides
$\dfrac{dy}{dx} = \dfrac{1}{\cos y}$	solve — still in y, so not finished
$\cos y = \sqrt{1-x^2}$	**convert**: triangle, or $\cos y = \sqrt{1-\sin^2 y}$
$\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1-x^2}}$	substitute



Step 5 in a picture. With $\sin y = x = \frac{x}{1}$, label the **opposite** side x and the **hypotenuse** 1. Pythagoras gives the adjacent side $\sqrt{1-x^2}$, and $\cos y$ is adjacent over hypotenuse.

The identity $\cos y = \sqrt{1-\sin^2 y}$ reaches the same place.

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4. (**3** pts) Find $\dfrac{dy}{dx}$ by implicit differentiation:

$$x \sin y = y$$

$\dfrac{dy}{dx} = \dfrac{\sin y}{1 - x \cos y}$

$\dfrac{d}{dx}[x \sin y] = \dfrac{d}{dx}[y]$	differentiate both sides
$\sin y + x \cos y \dfrac{dy}{dx} = \dfrac{dy}{dx}$	**Product Rule**; $\sin y$ needs the **Chain Rule**
$\sin y = \dfrac{dy}{dx} - x \cos y \dfrac{dy}{dx}$	gather the $\frac{dy}{dx}$ terms on one side
$\sin y = \dfrac{dy}{dx}(1 - x \cos y)$	factor
$\dfrac{dy}{dx} = \dfrac{\sin y}{1 - x \cos y}$	divide

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5. (**2** pts) Differentiate $y = 5^{x^2}$. You do not need to simplify.

$y' = 5^{x^2} (\ln 5)(2x)$

$\dfrac{d}{dx}[a^u] = a^u (\ln a) \dfrac{du}{dx}$	the any-base rule, with $a = 5,\ u = x^2$
$y' = 5^{x^2} (\ln 5)(2x)$	**Chain Rule**: $\frac{du}{dx} = 2x$

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6. (**3** pts) Use **logarithmic differentiation** to find $\dfrac{dy}{dx}$ for

$$y = x^{2x}, \qquad x > 0.$$

$\dfrac{dy}{dx} = x^{2x}(2 \ln x + 2)$

$\ln y = \ln\left(x^{2x}\right)$	take ln of both sides
$\ln y = 2x \ln x$	$\ln(A^p) = p \ln A$
$\dfrac{1}{y}\dfrac{dy}{dx} = 2 \ln x + 2x \cdot \dfrac{1}{x}$	differentiate; **Product Rule** on the right
$\dfrac{1}{y}\dfrac{dy}{dx} = 2 \ln x + 2$	simplify
$\dfrac{dy}{dx} = y(2 \ln x + 2) = x^{2x}(2 \ln x + 2)$	multiply by y, then put y back

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7. (**2** pts) Each edge of a cube is growing at 2 cm/s. Let s be the edge length and V the volume.

Write the relating equation and **differentiate both sides with respect to t. Do not substitute any numbers and do not solve.**

$V = s^3$ **and** $\dfrac{dV}{dt} = 3s^2 \dfrac{ds}{dt}$

$V = s^3$ the relating equation — true at **every** instant

$\dfrac{dV}{dt} = 3s^2 \dfrac{ds}{dt}$ **Chain Rule**: s moves, so it carries its own rate

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